

Digital Voter Information & Management System (VIMS)

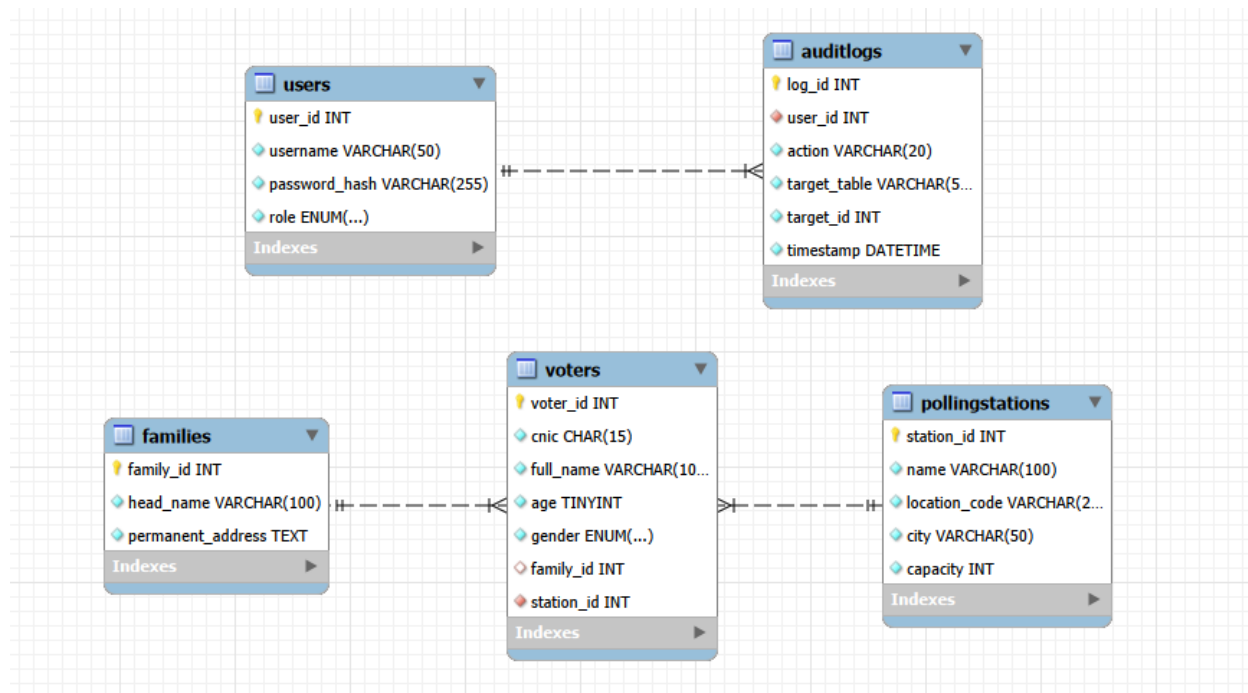
Suhaib Ahmad	Muhammad Rehan Khan
BS Software Engineering	BS Software Engineering
Group B	Group B

GitHub: <https://github.com/SuhaibAhmad0507/digiVIMS>

Version Control Document

Milestone	Version	Submission Date	Remarks
M1 — ERD & Relational Schema	v1.0	April 26, 2025	Changes Recommended
M1 — Add description with schema	v1.1	May 6, 2025	Schema descriptions added
M2 — ERD Design & Normalization	v2.0	May 2025	Approved by teacher
M3 — Dataset Preprocessing & Dataflow	v3.0	May 2025	Approved by teacher
M4 — Database Setup (DDL)	v4.0	May 2025	Approved by teacher
M5 — Data Population (DML)	v5.0	May 2025	Approved by teacher

Entity-Relationship Diagram (ERD)



GitHub Repository & Project Structure

Field	Details
Repository URL	https://github.com/SuhaibAhmad0507/digiVIMS
Group	Group B — Database Systems Lab, IMSciences
Members	Suhaib Ahmad Muhammad Rehan Khan (BS Software Engineering)

Repository Directory Layout

- **README.md** — Comprehensive project setup, deployment blueprint, and core dataflow details.
- **NORMALIZATION.md** — Exhaustive 1NF → 3NF walkthrough for all 5 relational entities.
- **data/** — Synthetic dataset CSV files: users, polling_stations, families, voters, audit_logs.
- **sql/ddl.sql** — Finalised DDL scripts with table structures, PK/FK allocations, and performance indexes.
- **sql/dml.sql** — DML scripts detailing LOAD DATA INFILE, safe structural updates, and transactional deletes.
- **sql/validation.sql** — Data verification suite: record counts, NULL checks, FK integrity checks, and core business queries.
- **generate_data.py** — Python/Faker data generation engine script to instantiate target records.

Commit History (Per-Milestone)

Milestone	Commit Message
M1	M1: Initial ERD and relational schema — v1.0
M1.1	M1.1: Add schema table descriptions — v1.1
M2	M2: Applied 2NF and 3NF normalization, updated ERD and schema
M3	M3: Synthetic data generated; dataflow documented
M4	M4: DDL scripts added, EER diagram verified
M5	M5: Data populated, validation queries added